Variable Selection Theorem for the Analysis of Covariance Model

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Sang-Hoo Yoon Jeong-Soo Park[‡]

Department of Statistics, Chonnam National University
Gwangju, Republic of Korea

Abstract

Variable selection theorem in the linear regression model is extended to the analysis of covariance model. When some of regression variables are omitted from the model, it reduces the variance of the estimators but introduces bias. Thus an appropriate balance between a biased model and one with small variances is recommended.

Keywords: Estimable function; generalized inverse; mean squared error; positive semi-definite matrix; reduced model.

1 Variable selection theorem in the regression model

To understand the theoretical basis for selecting appropriate explanatory variables when fitting a regression model, consider the following full and reduced models:

$$\text{[Full model]} \qquad y = X_p\beta_p + X_r\beta_r + \varepsilon, \tag{1.1}$$

$$\text{[Reduced model]} \qquad y = X_p\beta_p + \varepsilon. \tag{1.2}$$

Assume that the error has mean zero and covariance matrix $\sigma^2 I$. Let $\widehat{\beta}_p$ and $\widehat{\beta}_r$ denote the least-squares estimators under the full model, and let $\widetilde{\beta}_p$ denote the least-squares estimator of β_p under the reduced model. If the full model is true but the reduced model is fitted, that is, if the model is underspecified, the following results are known (Hocking, 1976).

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Theorem 1.1 (Hocking's variable selection theorem)**.**

1. $\widetilde{\beta}_p$ *is generally biased. It is unbiased in the special cases* $\beta_r = 0$ *or* $X_p^\top X_r = 0$.
2. $\mathrm{Var}(\widehat{\beta}_p) - \mathrm{Var}(\widetilde{\beta}_p)$ *is positive semidefinite.*
3. *If* $\mathrm{Var}(\widehat{\beta}_r) - \beta_r \beta_r^\top$ *is positive semidefinite, then* $\mathrm{MSE}(\widehat{\beta}_p) - \mathrm{MSE}(\widetilde{\beta}_p)$ *is also positive semidefinite.*
4. *For a specified value* x_0 *of the predictors,* $\mathrm{Var}(x_0^\top \widehat{\beta}) - \mathrm{Var}(x_{0p}^\top \widetilde{\beta}_p) \geq 0$.
5. *The variance estimator* $\widetilde{\sigma}^2$ *under the reduced model is upward biased.*

The theorem shows that fitting a reduced model introduces bias into coefficient estimation but decreases variance. In some circumstances, mean squared error (MSE) also decreases. This provides a theoretical rationale for variable elimination: when the appropriate positive-semidefiniteness condition holds, the reduced model may be preferable to the full model in terms of MSE. The theorem has appeared in many textbooks (e.g., Park, 1980; Rencher, 2000; Ravishanker and Dey, 2001; Seber and Lee, 2003).

In this paper, we extend Theorem 1.1 to an analysis of covariance (ANCOVA) model. Specifically, we investigate how estimators change when a reduced ANCOVA model is formed by omitting some regression variables. We show that the variances of both regression-coefficient estimators and analysis-of-variance effect estimators decrease, although biases are introduced, and that under certain conditions their MSEs decrease as well.

Related Korean studies include Kim and Kim (2004), Yoon and Song (2005), Hong et al. (2005), Choi (2006), and Park (2007). Related international studies include George (2000), Kadane and Lazar (2004), Yuan and Lin (2005), Gurka (2006), and Claeskens et al. (2006).

2 The analysis of covariance model and properties of its estimators

Analysis of covariance (ANCOVA) uses covariates to explain variation not accounted for by treatments, thereby improving experimental precision. It combines an analysis-of-variance model with a regression model. The design matrix for the analysis-of-variance component is generally not of full column rank, whereas the regression design matrix is assumed to have full column rank.

For n observations and $k = w + q$ explanatory variables, consider

$$y = X\tau + Z\beta + \varepsilon. \tag{2.1}$$

Here X is an $n \times w$ matrix that is not of full column rank; τ is a $w \times 1$ vector; Z is an $n \times q$ full-column-rank matrix with $q = p+r+1$; β is $q \times 1$; and $y = (Y_1, \ldots, Y_n)^\top$ is $n \times 1$. The columns of X associated with the analysis-of-variance terms are linearly independent of the columns of Z associated with the regression terms, in the sense required for estimation. The error ε has mean zero and covariance matrix $\sigma^2 I_n$.

The least-squares estimators of β and τ are (Ravishanker and Dey, 2001)

$$\widehat{\beta} = (Z^\top Q Z)^{-1} Z^\top Q y, \tag{2.2}$$

$$\widehat{\tau} = (X^\top X)^- X^\top (y - Z\widehat{\beta}), \tag{2.3}$$

where

$$Q = I - X(X^\top X)^- X^\top \tag{2.4}$$

is symmetric and idempotent. Here $(X^\top X)^-$ is a generalized inverse satisfying $(X^\top X)(X^\top X)^-(X^\top X) = X^\top X$.

Using $QX\tau = 0$, the regression estimator has expectation

$$E(\widehat{\beta}) = (Z^\top Q Z)^{-1} Z^\top Q (X\tau + Z\beta) = \beta. \tag{2.5}$$

For the treatment-effects estimator,

$$E(\widehat{\tau}) = (X^\top X)^- X^\top X \tau.$$

Because $(X^\top X)^- X^\top X$ cannot generally be removed, $\widehat{\tau}$ is not uniquely determined and need not estimate the parameter vector τ without ambiguity. However, for any estimable function $c^\top \tau$, we have $c^\top (X^\top X)^- X^\top X = c^\top$, and hence

$$E(c^\top \widehat{\tau}) = c^\top (X^\top X)^- X^\top X \tau = c^\top \tau. \tag{2.6}$$

The variance of $\widehat{\beta}$ is

$$\operatorname{Var}(\widehat{\beta}) = (Z^\top Q Z)^{-1} Z^\top Q Q^\top Z (Z^\top Q Z)^{-1} \sigma^2 \tag{1}$$

$$= (Z^\top Q Z)^{-1} \sigma^2. \tag{2.7}$$

Moreover,

$$\operatorname{Var}(c^\top \widehat{\tau}) = c^\top (X^\top X)^- X^\top \operatorname{Var}(y - Z\widehat{\beta}) X (X^\top X)^- c \tag{2}$$

$$= c^\top (X^\top X)^- X^\top \operatorname{Var}\{(I - WQ)y\} X (X^\top X)^- c \tag{3}$$

$$= c^\top (X^\top X)^- X^\top (I + W) X (X^\top X)^- c\, \sigma^2, \tag{2.8}$$

where the symmetric matrix W is

$$W = Z(Z^\top Q Z)^{-1} Z^\top. \tag{2.9}$$

The derivation uses $WQQ^\top W^\top = W$ and $c^\top (X^\top X)^- X^\top W Q X (X^\top X)^- c = 0$.

3 Estimators under a reduced analysis of covariance model

Partition $Z = (Z_p, Z_r)$ and rewrite model (2.1) as

$$\text{[Full model]} \qquad y = X\tau + Z_p\beta_p + Z_r\beta_r + \varepsilon. \tag{3.1}$$

Denote the least-squares estimators under the full model by $\widehat{\beta}_p$ and $\widehat{\beta}_r$. If Z_r is omitted, the reduced model is

$$\text{[Reduced model]} \qquad y = X\tau + Z_p\beta_p + \varepsilon. \tag{3.2}$$

The reduced-model estimators are

$$\widetilde{\beta}_p = (Z_p^\top Q Z_p)^{-1} Z_p^\top Q y, \tag{3.3}$$

$$\widetilde{\tau} = (X^\top X)^- X^\top (y - Z_p\widetilde{\beta}_p). \tag{3.4}$$

Suppose that (3.1) is true but (3.2) is fitted. Expectations are computed under the full model. Since $QX\tau = 0$,

$$E(\widetilde{\beta}_p) = (Z_p^\top Q Z_p)^{-1} Z_p^\top Q(X\tau + Z\beta) \tag{4}$$

$$= (Z_p^\top Q Z_p)^{-1} Z_p^\top Q(Z_p\beta_p + Z_r\beta_r) \tag{5}$$

$$= \beta_p + D\beta_r, \tag{3.5}$$

where

$$D = (Z_p^\top Q Z_p)^{-1} Z_p^\top Q Z_r. \tag{3.6}$$

Thus the bias is $D\beta_r$. Furthermore,

$$\mathrm{Var}(\widetilde{\beta}_p) = (Z_p^\top Q Z_p)^{-1} Z_p^\top Q Q^\top Z_p (Z_p^\top Q Z_p)^{-1}\sigma^2 \tag{6}$$

$$= (Z_p^\top Q Z_p)^{-1}\sigma^2, \tag{3.7}$$

and

$$\mathrm{MSE}(\widetilde{\beta}_p) = (Z_p^\top Q Z_p)^{-1}\sigma^2 + D\beta_r\beta_r^\top D^\top. \tag{3.8}$$

We next examine the effect of model misspecification on estimators of the analysis-of-variance effects. For an estimable function $c^\top\tau$, using $E(y) = X\tau + Z_p\beta_p + Z_r\beta_r$ yields

$$E(c^\top\widetilde{\tau}) = c^\top(X^\top X)^- X^\top E(y) - c^\top(X^\top X)^- X^\top Z_p E(\widetilde{\beta}_p) \tag{7}$$

$$= c^\top\tau - c^\top(X^\top X)^- X^\top(Z_p D - Z_r)\beta_r. \tag{3.9}$$

By an argument similar to that for (2.8),

$$\begin{aligned} \operatorname{Var}(c^\top \widetilde{\tau}) &= c^\top (X^\top X)^- X^\top \operatorname{Var}(y - Z_p \widetilde{\beta}_p) X (X^\top X)^- c \qquad (8) \\ &= c^\top (X^\top X)^- X^\top (I + W_p) X (X^\top X)^- c\, \sigma^2, \qquad (3.10) \end{aligned}$$

where

$$W_p = Z_p (Z_p^\top Q Z_p)^{-1} Z_p^\top. \qquad (3.11)$$

Combining (3.9) and (3.10), we obtain

$$\begin{aligned} \operatorname{MSE}(c^\top \widetilde{\tau}) = {} & c^\top (X^\top X)^- X^\top (I + W_p) X (X^\top X)^- c\, \sigma^2 \\ & + c^\top (X^\top X)^- X^\top (Z_p D - Z_r) \beta_r \beta_r^\top (Z_p D - Z_r)^\top X (X^\top X)^- c. \qquad (3.12) \end{aligned}$$

4 Extension of the variable selection theorem to ANCOVA

The preceding results yield the following extension of Theorem 1.1.

Theorem 4.1 (Variable selection theorem for ANCOVA)**.**

1. $\operatorname{Var}(\widehat{\beta}_p) - \operatorname{Var}(\widetilde{\beta}_p)$ *is positive semidefinite.*
2. $\operatorname{Var}(c^\top \widehat{\tau}) - \operatorname{Var}(c^\top \widetilde{\tau}) \geq 0$.
3. *If* $\operatorname{Var}(\widehat{\beta}_r) - \beta_r \beta_r^\top$ *is positive semidefinite, then (i)* $\operatorname{MSE}(\widehat{\beta}_p) - \operatorname{MSE}(\widetilde{\beta}_p)$ *is positive semidefinite; and (ii)* $\operatorname{MSE}(c^\top \widehat{\tau}) - \operatorname{MSE}(c^\top \widetilde{\tau}) \geq 0$.
4. $\widetilde{\sigma}^2$ *is upward biased.*

Before giving the proof, we establish some matrix identities. The block-inverse formula gives

$$\operatorname{Var}(\widehat{\beta}) = \sigma^2 (Z^\top Q Z)^{-1} = \sigma^2 \begin{pmatrix} Z_p^\top Q Z_p & Z_p^\top Q Z_r \\ Z_r^\top Q Z_p & Z_r^\top Q Z_r \end{pmatrix}^{-1} = \sigma^2 \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}. \qquad (4.1)$$

Using D in (3.6),

$$\begin{aligned} A_{11} &= (Z_p^\top Q Z_p)^{-1} + D A_{22} D^\top, \qquad (4.2) \\ A_{12} &= -D A_{22}, \qquad A_{21} = -A_{22} D^\top, \\ A_{22} &= (Z_r^\top Q Z_r - D^\top Z_p^\top Q Z_r)^{-1}. \qquad (4.3) \end{aligned}$$

From (2.9) and the block-inverse formula,

$$W = Z_p(Z_p^\top Q Z_p)^{-1} Z_p^\top + Z_p D A_{22} D^\top Z_p^\top + Z_r A_{21} Z_p^\top + Z_p A_{12} Z_r^\top + Z_r A_{22} Z_r^\top. \tag{4.4}$$

Consequently,

$$\begin{aligned} W - W_p &= Z_p D A_{22} D^\top Z_p^\top - Z_r A_{22} D^\top Z_p^\top - Z_p D A_{22} Z_r^\top + Z_r A_{22} Z_r^\top \\ &= (Z_p D - Z_r) A_{22} (Z_p D - Z_r)^\top. \end{aligned} \tag{4.5}$$

As $A_{22} = \mathrm{Var}(\widehat{\beta}_r)/\sigma^2$ is positive semidefinite, so is $W - W_p$.

Proof of Theorem 4.1.

1. By (4.1)–(4.2),

$$\mathrm{Var}(\widehat{\beta}_p) = \sigma^2 A_{11} = \sigma^2 (Z_p^\top Q Z_p)^{-1} + \sigma^2 D A_{22} D^\top. \tag{4.6}$$

Subtracting (3.7) gives

$$\mathrm{Var}(\widehat{\beta}_p) - \mathrm{Var}(\widetilde{\beta}_p) = \sigma^2 D A_{22} D^\top \succeq 0.$$

2. By (2.8) and (3.10),

$$\mathrm{Var}(c^\top \widehat{\tau}) - \mathrm{Var}(c^\top \widetilde{\tau}) = \sigma^2 c^\top (X^\top X)^- X^\top (W - W_p) X (X^\top X)^- c \geq 0,$$

since (4.5) shows that $W - W_p$ is positive semidefinite.

3(i). Using (2.5), (3.8), and (4.6),

$$\mathrm{MSE}(\widehat{\beta}_p) - \mathrm{MSE}(\widetilde{\beta}_p) = D(\sigma^2 A_{22} - \beta_r \beta_r^\top) D^\top. \tag{4.7}$$

The assertion follows because $\mathrm{Var}(\widehat{\beta}_r) = \sigma^2 A_{22}$.

3(ii). By (2.8), (3.12), and (4.5),

$$\begin{aligned} &\mathrm{MSE}(c^\top \widehat{\tau}) - \mathrm{MSE}(c^\top \widetilde{\tau}) \\ &= c^\top (X^\top X)^- X^\top \Big[(W - W_p)\sigma^2 - (Z_p D - Z_r) \beta_r \beta_r^\top (Z_p D - Z_r)^\top \Big] X (X^\top X)^- c \\ &= c^\top (X^\top X)^- X^\top (Z_p D - Z_r)(\sigma^2 A_{22} - \beta_r \beta_r^\top)(Z_p D - Z_r)^\top X (X^\top X)^- c \geq 0. \end{aligned}$$

4. Write $r_X = \mathrm{rank}(X)$.[1] The reduced-model estimator of σ^2 is

$$\widetilde{\sigma}^2 = \frac{(y - X\widetilde{\tau} - Z_p \widetilde{\beta}_p)^\top (y - X\widetilde{\tau} - Z_p \widetilde{\beta}_p)}{n - r_X - p - 1}. \tag{4.8}$$

[1]The original Korean article used $n - w - p - 1$ as the residual degrees of freedom in (4.8)–(4.11), with w denoting the number of columns of X, although X may be rank deficient. For mathematical correctness, the denominator is written here as $n - r_X - p - 1$, where $r_X = \mathrm{rank}(X)$. This is the only correction made to these formulas in the English translation.

Let $B = X(X^\top X)^- X^\top + QW_pQ$. The residual vector equals $(I - B)y$, so its residual sum of squares is $\mathrm{SSE} = y^\top (I - B)^\top (I - B)y$. Since $E(y) = X\tau + Z\beta$ and $\mathrm{Var}(y) = \sigma^2 I$,

$$\begin{aligned} E(\mathrm{SSE}) = \sigma^2 \operatorname{tr}\{(I - B)^\top (I - B)\} \\ + (X\tau + Z\beta)^\top (I - B)^\top (I - B)(X\tau + Z\beta). \end{aligned}$$

Using $I - B = Q(I - W_pQ)$ and the symmetry and idempotence of Q,

$$\begin{aligned} \operatorname{tr}\{(I - B)^\top (I - B)\} &= \operatorname{tr}(Q) - 2\operatorname{tr}(W_pQ) + \operatorname{tr}(W_pQW_pQ) \\ &= (n - r_X) - 2(p + 1) + (p + 1) \\ &= n - r_X - p - 1. \end{aligned}$$

Thus

$$E(\widetilde{\sigma}^2) = \sigma^2 + \frac{(X\tau + Z\beta)^\top (I - B)^\top (I - B)(X\tau + Z\beta)}{n - r_X - p - 1}. \tag{4.9}$$

The second term is nonnegative, proving upward bias. □

Finally, because $QX\tau = 0$ and $I - W_pQ$ is idempotent, the numerator of the second term of (4.9) can also be written as

$$\beta^\top Z^\top Q(I - W_pQ)Z\beta. \tag{4.10}$$

Partitioning $Z = (Z_p, Z_r)$ reduces this expression to $\beta_r^\top Z_r^\top Q(I - W_pQ)Z_r\beta_r$, or equivalently $\beta_r^\top Z_r^\top (I - B)Z_r\beta_r$. Therefore,

$$E(\widetilde{\sigma}^2) = \sigma^2 + \frac{\beta_r^\top Z_r^\top (I - B)Z_r\beta_r}{n - r_X - p - 1}. \tag{4.11}$$

5 Conclusion

It is known that fitting a reduced regression model introduces bias into the coefficient estimates but decreases their variance. This provides a theoretical basis for variable elimination: under suitable conditions, the reduced model may be preferable to the full model in terms of mean squared error. In this paper, we extended this result to the analysis of covariance (ANCOVA) model.

ANCOVA combines an analysis-of-variance model with regression. It uses covariates to explain variation not accounted for by treatment effects and can thereby improve experimental precision. Typically, the analysis-of-variance design matrix is not of full column rank, while the regression design matrix is of full column rank.

We examined the effects of omitting some regression variables from ANCOVA. For both regression coefficients and analysis-of-variance effects, the estimators may become biased while their variances decrease. Under certain conditions, their mean squared errors also decrease. Hence an appropriate model should balance bias against variance, or equivalently, aim to reduce mean squared error.

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